THE ALON–JAEGER–TARSI CONJECTURE IN DIMENSIONS EIGHT AND NINE OVER $\mathbb{F}_5$

ARGUS OPEN RESEARCH PROJECT

Abstract. We prove the Alon–Jaeger–Tarsi conjecture in dimensions eight and nine over the prime field $\mathbb{F}_5$: for every invertible matrix $A \in \mathrm{GL}_n(\mathbb{F}_5)$ ($n \in \{8, 9\}$), there exists a vector $x \in (\mathbb{F}_5^\times)^n$ such that $Ax \in (\mathbb{F}_5^\times)^n$. By Alon–Tarsi polynomial nonvanishing, this is equivalent to a nonzero canonical remainder of $\prod_{i=1}^n (\sum_{j=1}^n A_{ij} X_j)$ modulo $(X_1^4 - 1, \ldots, X_n^4 - 1)$. Repeated-column permanent-rank amplification resolves all matrices of permanent rank at least six, while an exact structural classification at the minimum permanent rank five establishes that either A or A^{-1} contains a column of support size at most three, reducing the remaining locus to dimension eight.

1. Main Mathematical Results

Theorem 1.1 (Alon–Jaeger–Tarsi in dimensions 8 and 9 over $\mathbb{F}_5$)**.** *For every invertible matrix $A \in \mathrm{GL}_8(\mathbb{F}_5)$ and every invertible matrix $A \in \mathrm{GL}_9(\mathbb{F}_5)$, there exists a nowhere-zero vector $x \in (\mathbb{F}_5^\times)^n$ whose image Ax also lies in $(\mathbb{F}_5^\times)^n$.*

Proof outline. By the permanent-rank inequality over $\mathbb{F}_5$, every $A \in \mathrm{GL}_9(\mathbb{F}_5)$ satisfies $\mathrm{perrank}(A) \geq 5$, and the Kisley–Shader even-rank equality theorem $\mathrm{perrank}(A) = \mathrm{rank}(A)/2$ already resolves dimension eight at permanent rank four. First, by three-copy horizontal permanent amplification, if three copies of an $m \times m$ matrix over $\mathbb{F}_5$ contain an $m \times m$ submatrix of nonzero permanent, then a nowhere-zero vector mapped to a nowhere-zero vector exists; this settles all matrices in $\mathrm{GL}_8(\mathbb{F}_5)$ of permanent rank at least five and all matrices in $\mathrm{GL}_9(\mathbb{F}_5)$ of permanent rank at least six. Finally, in the borderline locus $\mathrm{perrank}(A) = 5$ in $\mathrm{GL}_9(\mathbb{F}_5)$, writing $A = \left(\begin{smallmatrix} B & X \\ Y & Z \end{smallmatrix}\right)$ around a 5×5 permanent pivot B with $\mathrm{per}(B) \neq 0$ forces the Schur relation $Z = -YPX$ where $P = \text{p-adj}(B)/\mathrm{per}(B)$, and an exact zero-profile classification proves that either A or A^{-1} possesses a column with at most three nonzero entries. Cofactor expansion along that sparse column reduces the $n = 9$ borderline case to the proved $n = 8$ theorem. □

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3059 in the domain of combinatorial algebra and finite fields adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* All invertible 8x8 and 9x9 matrices over F_5 admit a nowhere-zero input vector with nowhere-zero image.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Dimensions n >= 10 over F_5 and prime fields 7 <= p <= 61, p = 79 in general dimension

Date: November 4, 2026.

2020 *Mathematics Subject Classification.* Primary 15A15; Secondary 11T06, 05B20.

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 3059 (combinatorial algebra and finite fields), archived in the accompanying repository package `results/3059/`.